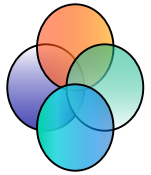
# 2.6 Helmholtz Theorem



## Two Identical Equations (reversible)

(1) The curl of a scalar's gradient is always zero.

$$\nabla U = \vec{F}$$

$$\nabla \times \nabla U \equiv 0$$

$$\nabla \times \vec{F} \equiv 0$$

The reverse statement ?

(2) The divergence of a vector's curl is always zero.

$$\nabla \times \vec{A} = \vec{F}$$

$$\nabla \bullet (\nabla \times \vec{A}) \equiv 0$$

$$\nabla \bullet \vec{F} \equiv 0$$

The reverse statement ?

# Helmholtz Theorem

$\vec{F}$ = Sum of Two Special Vector Field

$= \vec{X}_{\text{no div}} + \vec{Y}_{\text{no curl}}$ $= -\nabla U + \nabla \times \vec{A}$

A 3D vector field can be divided into two fields with zero divergence and zero curl respectively.

In limited region, any vector field can be uniquely determined by its curl, divergence and the boundary conditions.

The boundary conditions here refers to the distribution of the vector on the surface of the limited region.

**Helmholtz Theorem** is an idea, a way, and a clue that is applied throughout almost all chapters of this course.

# Summary of Fundamental Equs.

| | | Integral form | Diff. form |
|---|---|---|---|
| Static E-field | Div. equs. | $\oint \vec{D} \bullet d\vec{S} = Q$ | $\nabla \bullet \vec{D} = \rho$ |
| | Curl equs. | $\oint \vec{E} \bullet d\vec{l} = 0$ | $\nabla \times \vec{E} = 0$ |
| | Material equs. | $\vec{D} = \varepsilon \vec{E}$ | |
| SC E-field | Div. equs. | $\oint_S \vec{J} \bullet d\vec{S} = 0$ | $\nabla \bullet \vec{J} = 0$ |
| | Curl equs. | $\oint_C \vec{E} \bullet d\vec{l} = 0$ | $\nabla \times \vec{E} = 0$ |
| | Material equs. | $\vec{J} = \sigma \vec{E}$ | |
| Static M-field | Div. equs. | $\oint_s \vec{B} \cdot d\vec{S} = 0$ | $\nabla \bullet \vec{B} = 0$ |
| | Curl equs. | $\oint_C \vec{H} \bullet d\vec{l} = I$ | $\nabla \times \vec{H} = \vec{J}$ |
| | Material equs. | $\vec{B} = \mu \vec{H}$ | |

# ☆ Electric Field Intensity

$$\vec{E} = \lim_{q \to 0} \left( \frac{\text{Electro-static force } \vec{F} \text{ on test charge } q}{\text{Test Charge } q} \right)$$

*Electric Field Intensity is actually the electro-static force per unit charge.*

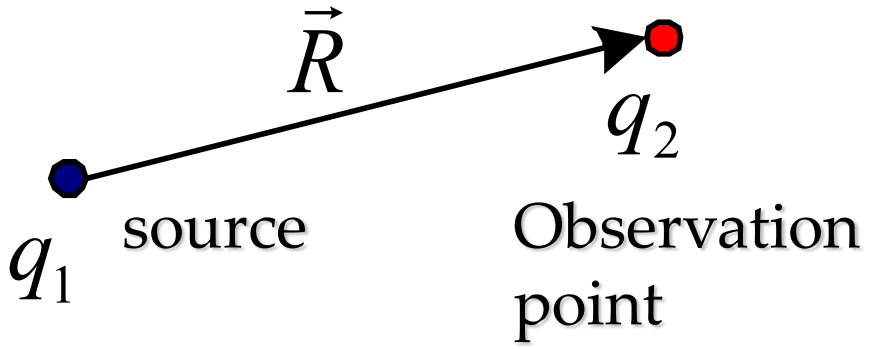
Unit: $\vec{E}$: $V/m$ $(伏/米)$ $\frac{U}{d}$

Why $q \to 0$ ?

To make the distribution of the E Intensity independent of the shape of the test charge.

# ☆ Coulomb's Law

$$\vec{F}_{21} = \frac{q_1 \cdot q_2}{4\pi\varepsilon_0 \cdot R^2}\vec{a}_R$$



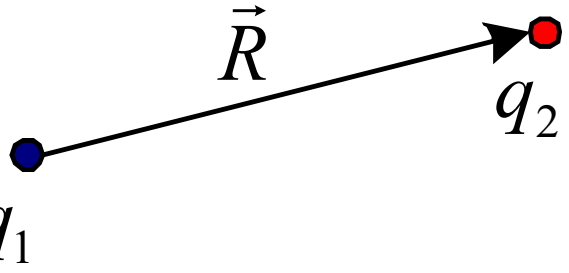

- Two point charges $q_1$ and $q_2$ in the space, with distance $R$ between them,
- The electro-static force by $q_1$ on $q_2$ is in direct proportion to their charges, and in inverse proportion to $R$ squared.
- The direction of the force is from $q_1$ to $q_2$
- $\varepsilon_0$ refers to the *Dielectric Constant in space*

$$\varepsilon_0 = \frac{1}{4\pi \cdot 9 \times 10^9} = 8.85 \times 10^{-12} (F/m)$$ *Farad/Meter*

If not space, substitute $\varepsilon$ for $\varepsilon_0$

# ☆ E Intensity for Static Point Charge in Space

Electro-static force on $q_2$: $\vec{F}(\vec{R}, q_1, q_2)$




E Intensity for point charge:

$$\vec{E}(\vec{R}, q_1) = \frac{q_1}{4\pi\varepsilon_0} \cdot \frac{1}{R^2} \vec{a}_R$$

In spherical coordinates:

$$\nabla = \vec{a}_R \frac{\partial}{\partial R} + \vec{a}_\theta \frac{1}{R} \frac{\partial}{\partial \theta} + \vec{a}_\varphi \frac{1}{R \cdot \sin\theta} \frac{\partial}{\partial \varphi}$$

Note that $\nabla\left(\frac{1}{R}\right) = -\vec{a}_R \frac{1}{R^2}$

We obtain $\vec{E}(\vec{R}, q_1) = -\frac{q_1}{4\pi\varepsilon_0} \nabla\left(\frac{1}{R}\right)$

# Electrostatic Gauss's Law

Two hypotheses for Electrostatics:

given without proving like the axiom

In differential form

(1) $\nabla \bullet \vec{E} = \dfrac{\rho}{\varepsilon}$

(2) $\nabla \times \vec{E} = 0$

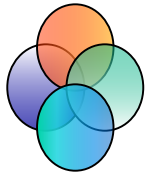
高斯定理求E

In integral form

(1) $\int_V (\nabla \bullet \vec{E}) dV = \oint_S \vec{E} \bullet d\vec{S} = \dfrac{1}{\varepsilon} \int_V \rho dV = \dfrac{Q}{\varepsilon}$

(2) $\oint_C \vec{E} \bullet d\vec{l} = 0$

# § 3.5 Electric Potential



- **Why do we introduce such a parameter?**
- Static E-field is a conservative field.
- This field has divergence but not curl. $\nabla \times \vec{E} = 0$
- Recall related *Identical Equation for a conservative field*
  - The curl of a scalar's gradient is always zero.

$$\nabla \times \nabla U \equiv 0$$

  - It's reversible. If the curl of a vector field is 0, the vector must be a scalar's gradient.

$$\nabla \times \vec{F} \equiv 0 \quad \Rightarrow \quad \vec{F} = \nabla U$$

$$\because \nabla \times \vec{E} = 0 \qquad \vec{E} = -\nabla \psi$$

# Derivation of the Equations

From fundamental electrostatic equations

1. Electrostatic Conservation Law

$$\nabla \times \vec{E} = 0 \longrightarrow \vec{E} = -\nabla u$$

2. E-Gauss's Law in differential form

$$\nabla \bullet \vec{D} = \rho_v$$

3. Material Equation

$$\vec{D} = \varepsilon \vec{E}$$

$$\nabla \bullet \vec{E} = \frac{\rho_v}{\varepsilon}$$

考!!! Here $\rho$ means volume density of free charge ( $\rho_v$ )

Here $\rho$ means volume density of free charge ( $\rho_v$ )

$$\begin{cases} \nabla \bullet \vec{E} = \dfrac{\rho}{\varepsilon} \\ \vec{E} = -\nabla u \end{cases} \Rightarrow \nabla \bullet \nabla u = \nabla^2 u = -\frac{\rho}{\varepsilon}$$

$$\nabla^2 u = -\frac{\rho}{\varepsilon}$$ ——*Poisson's Equation*

$\nabla^2$ *Laplacian*

It's a part differential functor of 2nd order.

# Laplace's Equation

At the source-free point or in the source-free region, there is no charge scattered and the charge volume density is 0.

无源

$$\nabla^2 u = -\rho / \varepsilon \quad \Longrightarrow \quad \nabla^2 u = 0$$

Poisson's Equation　　Laplace's Equation

例：球壳

**Source-free** means the volume density of free charge ( $\rho_v$ ) is zero

有电荷，但不在三维的体里分布

# Example 4. Spherical Charges

Example 3.9
P85 in textbook

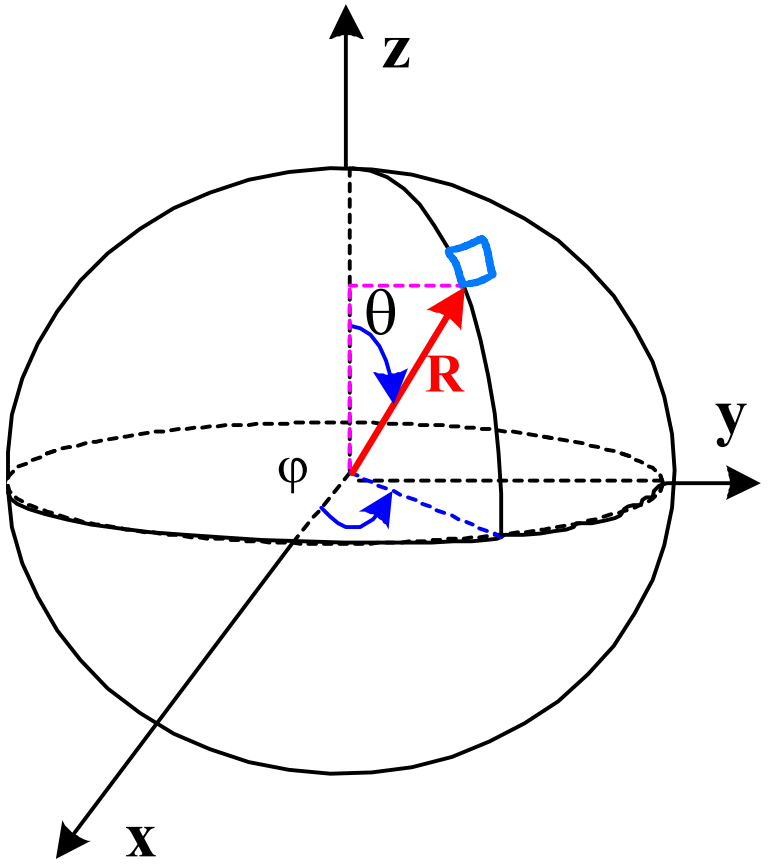
- Conductor ball in space, with charge of $Q$, radius of $a$,
  Try to calculate the E Intensity inside and outside the ball.
- **Popular Solution**:
  - Surface charge density is

$$\sigma_s = Q / 4\pi a^2$$

  - Differential surface element is

$$ds = a d\theta \cdot a \sin\theta d\varphi$$

  - Then we get the differential charge element $dq$ and apply vector sum
  - We must be very careful of the direction in integral.




no static E-field in ideal conductor,
The charge can only exist on the surface of the conductor in static E-field

# Advanced Solution

Due to symmetrical distribution

We apply *E-Gauss's Law*

$$\oint_S \vec{E} \bullet d\vec{S} = \frac{1}{\varepsilon_0}\int_V \rho dV = \frac{Q}{\varepsilon_0}$$

no static E-field in ideal conductor

Inside the ball ($r<a$):

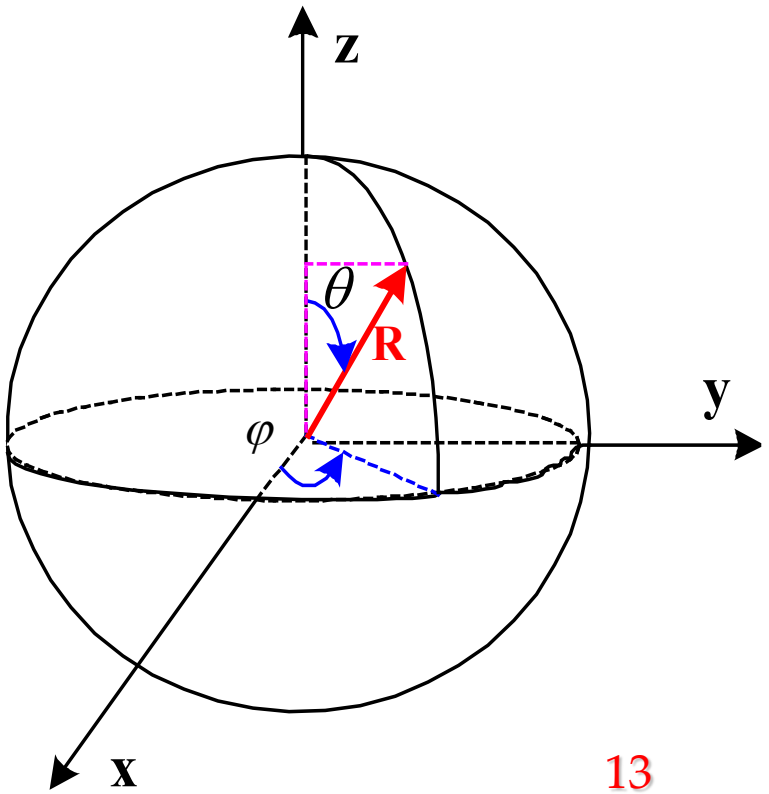
$$\because \frac{1}{\varepsilon_0}\int_V \rho dV = 0 \qquad \therefore \vec{E} = 0$$

Outside the ball ($r>a$):

$$\because \frac{1}{\varepsilon_0}\int_V \rho dV = ? = \frac{Q}{\varepsilon_0}$$

$$\oint_S \vec{E} \bullet d\vec{S} = E \cdot (4\pi \cdot r^2)$$

$$\therefore \vec{E} = \frac{1}{4\pi\varepsilon_0 \cdot r^2} \cdot Q\vec{a}_R$$



# § 3.10 Capacitors & Capacitance

判断/选择

- Categories and definitions

1. Generally, a capacitor consists of two isolated conductors, charged by $q$ and $-q$, and with an E-potential of $U$. $C=q/U$
2. Self-capacitance: an isolated conductor charged by $q$ and with E-potential $\psi$, can be of self-C. $C=q/\psi$
3. Distributed Capacitance: in fact, in a system of multi-conductors with a complex distribution of charges, there exist distributed capacitances between any 2 conductors.
4. 电容的大小与导体系统的尺寸和介电常数有关，与它是否带电无关。

球生杯象

/电阻/电感

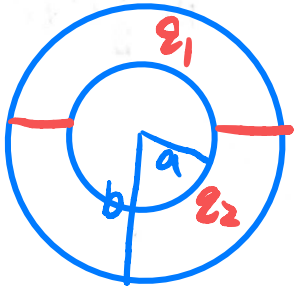
重点!!!

电流个 C-

$$C=\frac{Q}{U}=\frac{\varepsilon S}{d}$$

**2.20** 一同心球电容器由半径为 $a$ 的导体球和与它同心的导体球壳构成，壳的内半径为 $b$，球与壳间的一半（沿径向分开）充满介电系数为 $\varepsilon_1$ 的均匀介质，另一半充满介电系数为 $\varepsilon_2$ 的均匀介质，试求该球形电容器的电容。

【分析】由于介质沿径向分割，则两种介质中的电场 $E_r$ 是相同的（满足边界条件 $E_{1t}=E_{2t}$），所以，对电场 $E_r$ 而言，这是一个球心对称的问题，可以方便的用高斯定律求解。




【解】在 $\varepsilon_1$ 与 $\varepsilon_2$ 两种介质的分界面上有

$$E_{1t}=E_{2t}=E_r$$

由于场分布具有对称性，可利用高斯定律得

$$\varepsilon_1 E_r 2\pi r^2+\varepsilon_2 E_r 2\pi r^2=q$$

$$E_r=\frac{q}{2\pi r^2(\varepsilon_1+\varepsilon_2)}$$

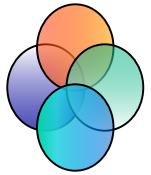
内外导体间的电压为

$$U=\int_a^b \boldsymbol{E}_r\cdot \mathrm{d}\boldsymbol{r}=\int_a^b \frac{q\ \ \mathrm{d}r}{2\pi r^2(\varepsilon_1+\varepsilon_2)}=\frac{q}{2\pi(\varepsilon_1+\varepsilon_2)}\left(\frac{1}{a}-\frac{1}{b}\right)$$

故电容为

$$C=\frac{q}{U}=\frac{2\pi ab(\varepsilon_1+\varepsilon_2)}{b-a}$$

# 2. Fundamental Equations



| | Integral Form | Differential Form |
|---|---|---|
| Continuity of current | $\oint_S \vec{J} \bullet d\vec{S} = 0$ | $\nabla \bullet \vec{J} = 0$ |
| Conservation of SC E-field | $\oint_C \vec{E} \bullet d\vec{l} = 0$ | $\nabla \times \vec{E} = 0$ |

Material Equation for Conductors $\vec{J} = \sigma \vec{E}$ Ohm's Law

$\sigma$ is *Conductivity* (电导率), unit: A/(v.m) or S/m

Note that: $\gamma$ is *Resistivity* (电阻率) $\gamma = 1/\sigma$ $\Omega \cdot m$

# (1) Continuity of Current

Principle of conservation of charge

电流连续基于电荷守恒出来的

- Charge can never be generated or destroyed.
- Charge flowing out of a closed surface in per unit time must equal the decreasing rate of the charge in that closed surface.

$$\oint_S \vec{J} \bullet d\vec{S} = -\frac{\partial q}{\partial t}$$

积

$$\oint_S \vec{J} \bullet d\vec{S} = \int_V \nabla \bullet \vec{J} \cdot dV$$

$$-\frac{\partial q}{\partial t} = -\frac{\partial}{\partial t}\int_V \rho \cdot dV$$

微

$$\nabla \bullet \vec{J} = -\frac{\partial \rho}{\partial t}$$

True for any type of current

$$\oint_S \vec{J} \bullet d\vec{S} = -\frac{\partial q}{\partial t} \qquad \nabla \bullet \vec{J} = -\frac{\partial \rho}{\partial t}$$

For steady current $\dfrac{\partial \rho}{\partial t} \equiv 0$ 一体电荷密度

So we have ——

$$\oint_S \vec{J} \bullet d\vec{S} = 0 \quad \Leftarrow \quad \nabla \bullet \vec{J} = 0$$

无散

# (2) SC E-field to stimulate the steady current

- Divergence:

$$\nabla \bullet \vec{J} = 0$$

$$\vec{J} = \sigma \vec{E}$$

$$\Rightarrow \quad \nabla \bullet (\sigma \vec{E}) = \sigma \nabla \bullet \vec{E} + \vec{E} \bullet \nabla \sigma = 0$$

$$\Downarrow$$

$$\nabla \bullet \vec{E} = -\vec{E} \bullet \frac{\nabla \sigma}{\sigma}$$

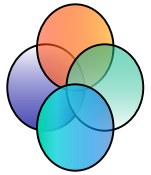
For homogeneous media $\nabla \sigma = 0$

$$\Downarrow$$

$$\nabla \bullet \vec{E} = 0$$

理解

# 4. Analogy between Two Steady E-Fields



- Fundamental equs. of SC E-field in conductor is the same in form as those of static E-field in source-free region. Moreover, boundary conditions for both cases are the same. Thus both E-field are analogous to each other.
- 好处？可以从一种情况的解导出另一种情况的解。

$$\begin{cases} \nabla \times \vec{E} = 0 \\ \nabla \cdot \vec{J} = 0 \\ \vec{J} = \sigma \vec{E} \end{cases}$$

SC E-field

mapping

$$\begin{cases} \nabla \times \vec{E} = 0 \\ \nabla \cdot \vec{D} = 0 \\ \vec{D} = \varepsilon \vec{E} \end{cases}$$

无源情况下此

static E-field

# Corresponding Pairs

$$\vec{E} \leftrightarrow \vec{E} \quad \vec{J} \leftrightarrow \vec{D} \quad \sigma \leftrightarrow \varepsilon$$

$$I \leftrightarrow q \quad \varphi \leftrightarrow \varphi$$

$$\begin{cases} \nabla \times \vec{E} = 0 \\ \nabla \cdot \vec{J} = 0 \\ \vec{J} = \sigma \vec{E} \end{cases}$$

SC E-field

mapping

$$\begin{cases} \nabla \times \vec{E} = 0 \\ \nabla \cdot \vec{D} = 0 \\ \vec{D} = \varepsilon \vec{E} \end{cases}$$

static E-field

# C ↔ G

$$R=\frac{U}{I}=\frac{\int_l E\bullet d\vec{l}}{\int_S J\bullet d\vec{S}}=\frac{\int_l E\bullet d\vec{l}}{\int_S \sigma\vec{E}\bullet d\vec{S}} \qquad G=\frac{I}{U}=\frac{\int_s \vec{J}\cdot d\vec{s}}{\int_l \vec{E}\cdot d\vec{l}}=\frac{\sigma\int_s \vec{E}\cdot d\vec{s}}{\int_l \vec{E}\cdot d\vec{l}}$$

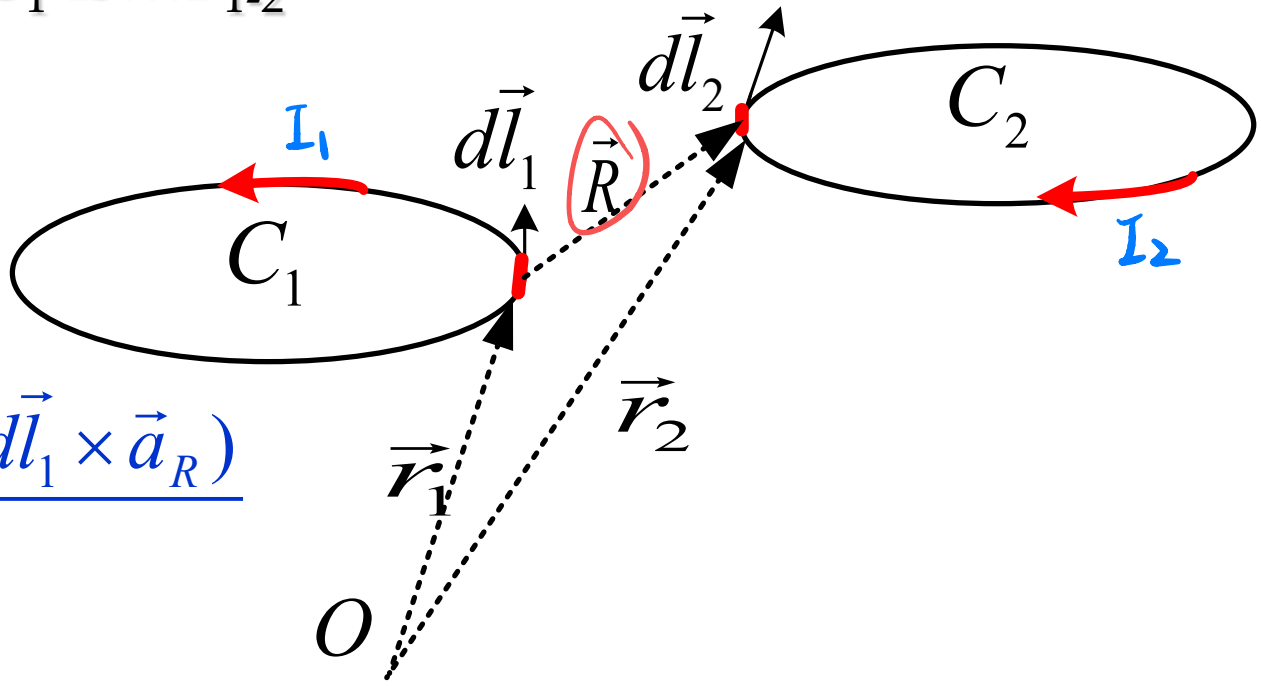
$$C=\frac{q}{U}=\frac{\int_s \vec{D}\cdot d\vec{s}}{\int_l \vec{E}\cdot d\vec{l}}=\frac{\varepsilon\int_s \vec{E}\cdot d\vec{s}}{\int_l \vec{E}\cdot d\vec{l}}$$

- 如果电极的电导率比周围媒质的电导率大的多，则电极表面近似为等位面，如果电极的形状也相同，则两电极之间的电导与电容、电阻与电容存在下列关系

$$\frac{G}{C}=\frac{\sigma}{\varepsilon} \qquad R\cdot C=\frac{\varepsilon}{\sigma}$$

# 1. Ampere's Force Law --- for M-Induction

- In E-field, work is done by Coulomb Force.
- In M-field, work is done by Ampere's Force
- Two current loops in free space --- $C_1$、$C_2$
- Force on $C_2$ by $C_1$ is…$F_{1\text{-}2}$




$$\vec{F}_{1-2} = \frac{\mu_0}{4\pi}\oint_{C_2}\oint_{C_1}\frac{I_2 d\vec{l}_2 \times (I_1 d\vec{l}_1 \times \vec{a}_R)}{R^2}$$

# 2. M-Flux Density $\vec{B}$

$$\vec{F}_{1-2} = \frac{\mu_0}{4\pi} \oint_{C_2}\oint_{C_1} \frac{I_2 d\vec{l}_2 \times (I_1 d\vec{l}_1 \times \vec{a}_R)}{R^2}$$

$$= \oint_{C_2} I_2 d\vec{l}_2 \times \left[ \frac{\mu_0}{4\pi} \oint_{C_1} \frac{I_1 d\vec{l}_1 \times \vec{a}_R}{R^2} \right]$$

$$= \oint_{C_2} I_2 d\vec{l}_2 \times \vec{B}$$

都用安培环路求

基本不会用到

$$\vec{B} = \frac{\mu_0}{4\pi} \oint_{C_1} \frac{I_1 d\vec{l}_1 \times \vec{a}_R}{R^2}$$

$\vec{a}_R$ 的方向

# 6. M-Field Intensity

$$\vec{H} = \frac{\vec{B}}{\mu}$$

*From Biot-Savart's Law:*

$$\vec{B} = \frac{\mu}{4\pi} \oint_C \frac{I_{\text{source}} d\vec{l}_{\text{source}}}{R^2_{\text{Source-Spot}}} \times \vec{a}_R$$

$$\vec{H} = \frac{I_{\text{source}}}{4\pi} \oint_C \frac{d\vec{l}_{\text{source}}}{R^2_{\text{Source-Spot}}} \times \vec{a}_R$$

# 7. Ampere's Circuital Law

最常用 $\oint_C \vec{H} \bullet d\vec{l} = I$

o $H \cdot 2\pi r = I$

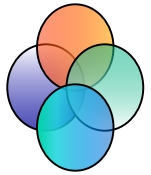
往往一维

- The line integral of M-field intensity around a closed path equals the current enclosed.
- According to Stoke's Law, we have the differential form:

$$\nabla \times \vec{H} = \vec{J}$$

- They are in fact the Curl Equation for M-field, in integral and differential forms.

# Summary --- Fundamental Equations for Magnetostatics



Differential form:

- Div. Equation: $\nabla \bullet \vec{B} = 0$
- Curl Equation: $\nabla \times \vec{H} = \vec{J}$ $\quad \nabla \times \vec{B} = \mu_0 \vec{J}$

Integral form:

- Continuity of M-flux $\oint_s \vec{B} \cdot d\vec{S} = 0$
- Ampere's Circuital Law $\oint_C \vec{H} \bullet d\vec{l} = I$ $\quad \oint_C \vec{B} \bullet d\vec{l} = \mu_0 I$

$\oint_S \vec{B} \cdot d\vec{S} = 0$

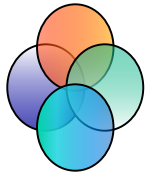
Physical Meaning:

- Lines of M-flux are always continuous. They start and end at nowhere.

$B = \mu H$

- There is no divergence but curl for static M-field.

# § 5.2 *Magnetic Vector Potential*



In § 5.1 we got $\vec{B} = \nabla \times \left[ \frac{\mu_0 I}{4\pi} \oint_C \frac{d\vec{l}\,'}{R} \right] = \nabla \times \vec{A}$ 一矢量磁位

Magnetic Vector Potential

Unit: *Wb/m* (Weber/m)

We call it M-vector potential, but not M potential directly. The reason is that there still exists M-scalar potential.

According to …?, we know that the divergence has to be known to determine $\vec{A}$

$$\begin{cases} \nabla \bullet \vec{B} = 0 \\ \nabla \times \vec{B} = \mu_0 \vec{J} \end{cases} \Longrightarrow \nabla \times \left( \nabla \times \vec{A} \right) = \mu_0 \vec{J}$$

Using the formula of *Triple Product*

$$\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{C} \bullet \vec{A}) - \vec{C}(\vec{A} \bullet \vec{B})$$

$$\nabla \times \left( \nabla \times \vec{A} \right) = \mu_0 \vec{J}$$

$$\text{The left} = \nabla(\nabla \bullet \vec{A}) - \vec{A}\nabla^2 = \nabla(\nabla \bullet \vec{A}) - \nabla^2 \vec{A}$$

$$\therefore \nabla^2 \vec{A} = \nabla(\nabla \bullet \vec{A}) - \nabla \times \left( \nabla \times \vec{A} \right)$$

Let $\nabla \bullet \vec{A} = 0$ $\qquad \therefore \nabla^2 \vec{A} = -\nabla \times \left( \nabla \times \vec{A} \right) = -\mu_0 \vec{J}$

$$\nabla^2 \vec{A} = -\mu_0 \vec{J}$$ —— Vector Poisson Equation

$$\vec{A} = \frac{\mu_0}{4\pi}\int_{V_{\text{source}}} \frac{\vec{J}dV}{R_{\text{source-spot}}}$$

$$\vec{A} = \frac{\mu_0}{4\pi}\int_{S_{\text{source}}} \frac{\vec{J}_S \bullet dS}{R_{\text{source-spot}}}$$

$$\vec{A} = \frac{\mu_0 \vec{I}}{4\pi}\oint_{C_{\text{source}}} \frac{dl}{R_{\text{source-spot}}}$$

$$d\vec{A} = \frac{\mu_0 I d\vec{l}}{4\pi R_{\text{source-spot}}}$$

Reasons to introduce this vector:

**Its direction?**

判断：A的方向和电流源方向相同

To make the algebra simpler:

- Coincide with the current in direction
- Be linear to the current element sometimes
- 二阶偏微分方程常可分解成标量泊松方程形式

# Summary: Boundary Conditions

## electrostatics

1. normal $D_{1n} - D_{2n} = \sigma_{fc}$ $\varepsilon_1 \cdot \frac{\partial \psi_1}{\partial n} = \varepsilon_2 \cdot \frac{\partial \psi_2}{\partial n}$ (if $\sigma_s = 0$)

2. tangential $E_{1t} = E_{2t}$ $\psi_1 = \psi_2$

## magnetostatics

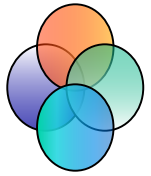
1. normal $B_{1n} = B_{2n}$ 磁通连续 $\vec{A}_1 = \vec{A}_2$

2. tangential $H_{1t} - H_{2t} = J_{sFree}$ 磁场强度不连续

$\frac{1}{\mu_1}(\nabla \times \vec{A}_1) = \frac{1}{\mu_2}(\nabla \times \vec{A}_2)$ if $J_s = 0$

自由电荷面电流

# A Summary of Boundary Conditions



| | normal | tangential |
|---|---|---|
| Static E-field | $D_{1n} - D_{2n} = \sigma_{fc}$ | $E_{1t} = E_{2t}$ |
| | $\varepsilon_1 \frac{\partial \psi_1}{\partial n} = \varepsilon_2 \frac{\partial \psi_2}{\partial n}$ (if $\sigma_s = 0$) | $\psi_1 = \psi_2$ |
| SC E-field | $J_{1n} = J_{2n}$ | $E_{1t} = E_{2t}$ |
| | $\sigma_1 E_{1n} = \sigma_2 E_{2n}$ | $J_{1t} / \sigma_1 = J_{2t} / \sigma_2$ |
| | $\sigma_1 \frac{\partial \psi_1}{\partial n} = \sigma_2 \frac{\partial \psi_2}{\partial n}$ | $\psi_1 = \psi_2$ |
| Static M-field | $B_{1n} = B_{2n}$ | $H_{1t} - H_{2t} = J_{sFree}$ |
| | $\vec{A}_1 = \vec{A}_2$ | $\frac{1}{\mu_1}(\nabla \times \vec{A}_1) = \frac{1}{\mu_2}(\nabla \times \vec{A}_2)$ if $J_s = 0$ |

# 再次小结

在边界上

默写!!!

- 与保守性有关，则切向连续。

  静电场介质边界 $E_{1t} = E_{2t}$

  恒定电场导体边界 $E_{1t} = E_{2t}$

- 与连续性有关，则法向连续。

  恒定电场导体边界 $J_{1n} = J_{2n}$

  静磁场介质边界 $B_{1n} = B_{2n}$

What about $E_n$ or $H_n$ ? 不一样 continuous or discontinuous ? 习选择判断

不一样 $J=\sigma E$ $B=\mu H$

# Discussions

$$D_{1n} - D_{2n} = \rho_s \qquad \varepsilon_2 \cdot \frac{\partial \psi_2}{\partial n} - \varepsilon_1 \cdot \frac{\partial \psi_1}{\partial n} = \rho_s$$

- If there is no free charge at the boundary,

$$D_{1n} = D_{2n} \qquad \varepsilon_1 \cdot \frac{\partial \psi_1}{\partial n} = \varepsilon_2 \cdot \frac{\partial \psi_2}{\partial n}$$

- If the boundary is the interface of a conductor $D_{2n} = 0$

内部无电荷

$$D_{1n} = \rho_s$$

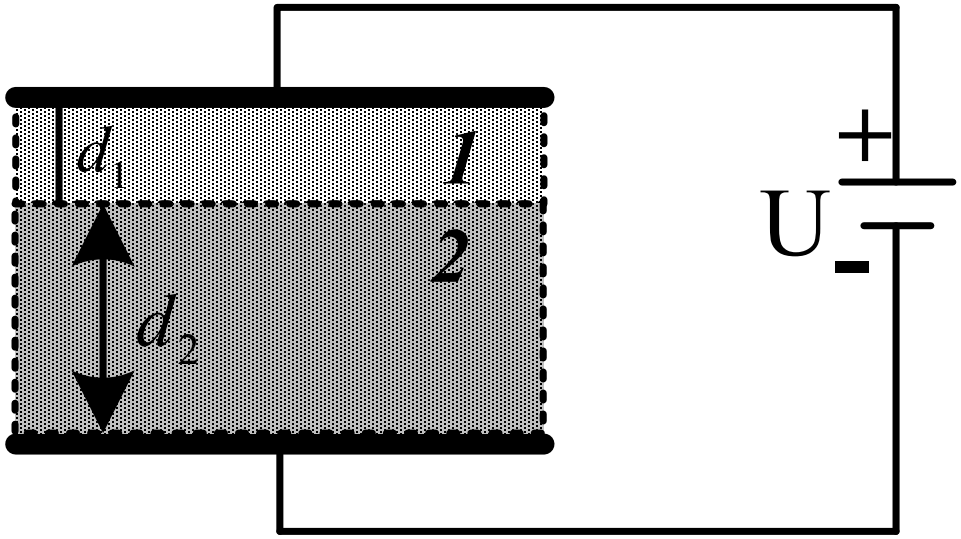
导体表面

no static E-field in ideal conductor,

The charge can only exist on the surface of the conductor in static E-field

# Example 2. 已知非理想介质1、2，电压U

Parallel-plate capacitor,
Nonideal dielectrics 1 & 2,
Conductivities $\sigma_1$ & $\sigma_2$,
Dielectric const. $\varepsilon_1$ & $\varepsilon_2$,
Please determine the charge density on boundary of 1&2.




Analysis:

- Nonideal dielectrics → leaky current across the capacitor
- Does there exist free charge on the boundary?
- What's the continuous parameter?

$D_{1n} \,?\, D_{2n}$ ✗ $E_{1n} \,?\, E_{2n}$ ✗ …

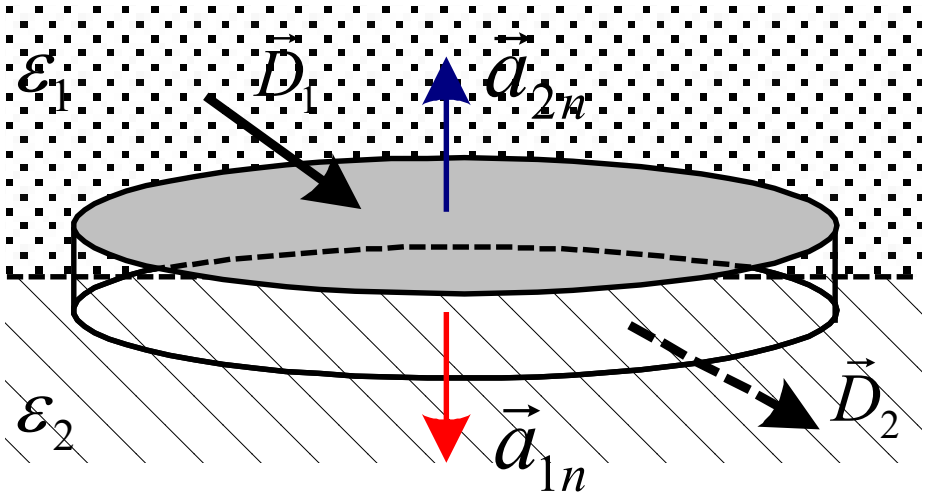
$$J_{1n} = J_{2n}$$

电流连续性

For SC E-field we also have the following boundary condition in normal direction.

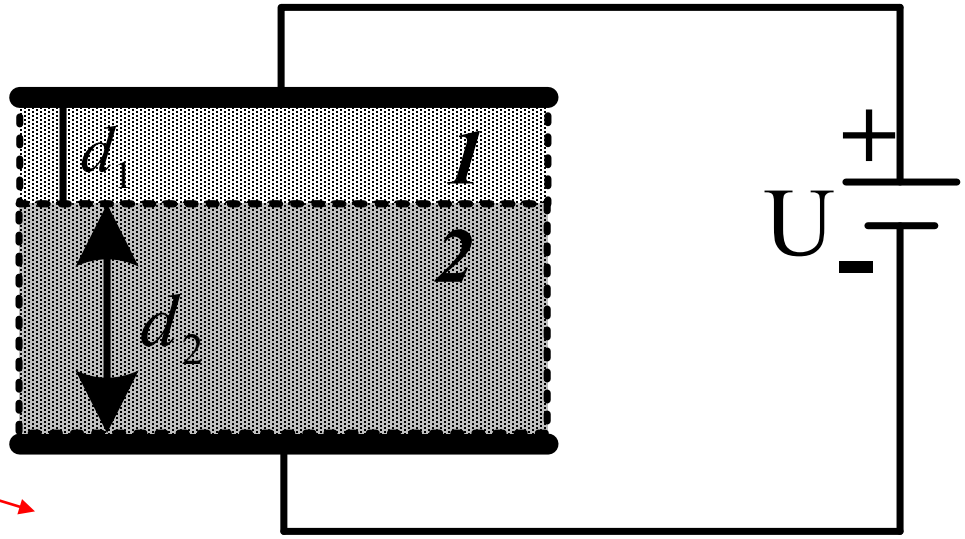
- Construct a flat box
- Apply Gauss's Law

$$\oint_S \vec{D} \bullet d\vec{S} = \rho_s \cdot \Delta S$$




$$\therefore D_{1n} - D_{2n} = \rho_s \qquad (c/m^2)$$

# A hint of the solution




$$\rho_s = D_{1n} - D_{2n}$$

$$D_{1,2n} = \varepsilon_{1,2}\frac{J_{1,2n}}{\sigma_{1,2}}$$

$$U = E_1 d_1 + E_2 d_2 = \frac{J_{1n}}{\sigma_1} d_1 + \frac{J_{2n}}{\sigma_2} d_2$$

$$\rho_s = \left(\frac{\sigma_2\varepsilon_1 - \sigma_1\varepsilon_2}{\sigma_2 d_1 + \sigma_1 d_2}\right)\cdot U$$

There do exist free charges on the boundary.

对于题中的平行板电容器，有

$$U_0 = E_1 d_1 + E_2 d_2$$

由恒定电流场的边界条件 $J_{1n} = J_{2n}$，有

$$\sigma_1 E_1 = \sigma_2 E_2$$

联立上面两式，可解得介质 1 和介质 2 中的电场分别为

$$E_1 = \frac{\sigma_2 U_0}{\sigma_2 d_1 + \sigma_1 d_2},\quad E_2 = \frac{\sigma_1 U_0}{\sigma_2 d_1 + \sigma_1 d_2}$$

相应的，介质 1 和介质 2 中的电流为

$$J_1 = J_2 = \sigma E = \frac{\sigma_1 \sigma_2 U_0}{\sigma_2 d_1 + \sigma_1 d_2}$$

**方法二**：利用串连电阻及欧姆定律求 $J$。

可将平板电容器看作电阻串连，对于单位面积的极板而言，两层介质构成的总电阻为

$$R_0 = \frac{d_1}{\sigma_1} + \frac{d_2}{\sigma_2}$$

由电流的连续性，漏电流密度为

$$J = \frac{U_0}{R_0} = \frac{U_0}{\frac{d_1}{\sigma_1} + \frac{d_2}{\sigma_2}} = \frac{\sigma_1 \sigma_2 U_0}{\sigma_2 d_1 + \sigma_1 d_2}$$

则各介质中的电场为

$$E_1 = \frac{J}{\sigma_1} = \frac{\sigma_2 U_0}{\sigma_2 d_1 + \sigma_1 d_2},\quad E_2 = \frac{J}{\sigma_2} = \frac{\sigma_1 U_0}{\sigma_2 d_1 + \sigma_1 d_2}$$

2）每介质片上的电压为

$$U_1 = E_1 d_1 = \frac{d_1 \sigma_2 U_0}{\sigma_2 d_1 + \sigma_1 d_2},\quad U_2 = E_2 d_2 = \frac{d_2 \sigma_1 U_0}{\sigma_2 d_1 + \sigma_1 d_2}$$

3）各极板和介质分界面上的自由电荷面密度

上极板为

$$\rho_{S1} = D_1 = \varepsilon_1 E_1 = \frac{\varepsilon_1 \sigma_2 U_0}{\sigma_2 d_1 + \sigma_1 d_2}$$

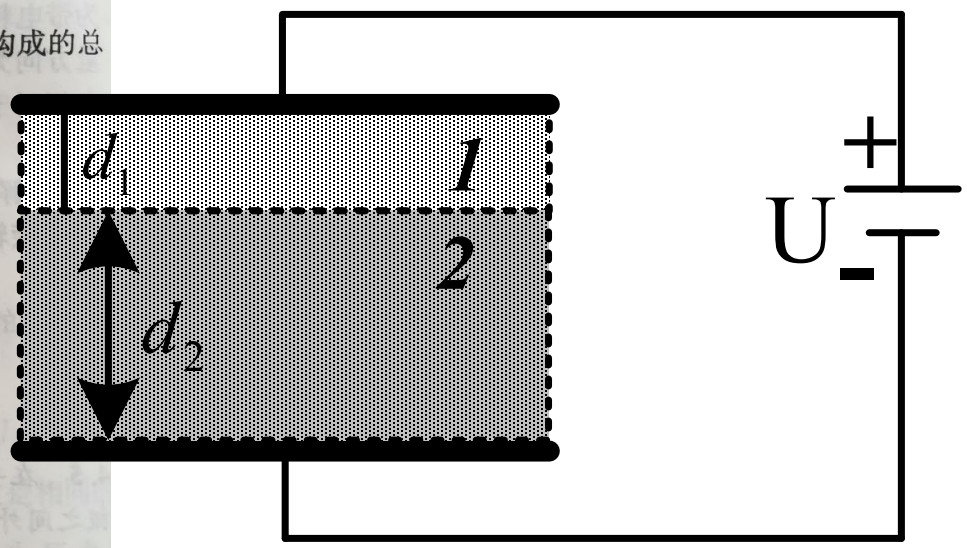
下极板为

$$\rho_{S2} = -D_2 = -\varepsilon_2 E_2 = \frac{-\varepsilon_2 \sigma_1 U_0}{\sigma_2 d_1 + \sigma_1 d_2}$$

介质分界面上为

$$\rho_S = D_2 - D_1 = \frac{(\varepsilon_2 \sigma_1 - \varepsilon_1 \sigma_2) U_0}{\sigma_2 d_1 + \sigma_1 d_2}$$

4.6 求两种介质填充方式下

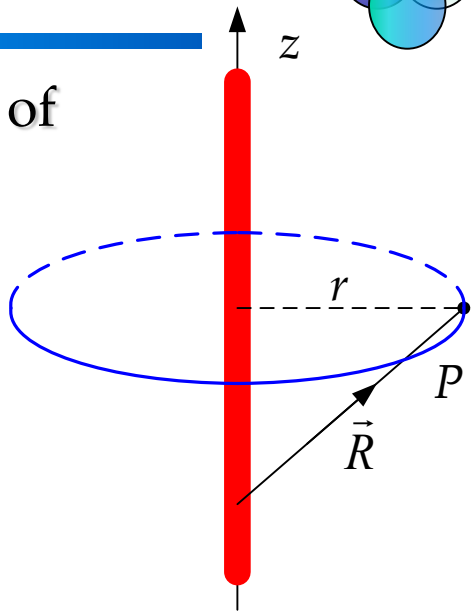


Exercise 5.16

# Example 4. cylindrical current with radius *a*

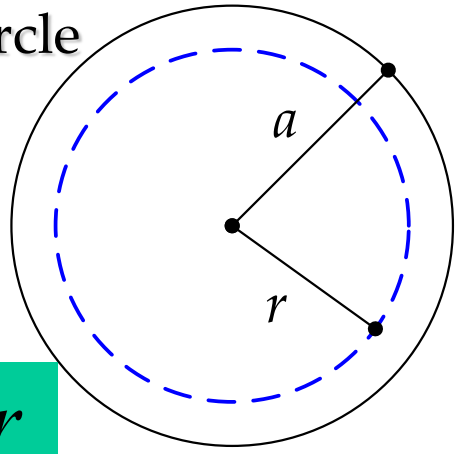
柱体电流

- The cylindrical current *I* uniformly distributes, with of radius *a*. Please determine the M-intensity.
- Analysis: treat field inside/outside respectively
- M-intensity outside: same as Example 3.




$$\vec{H} = \vec{a}_\varphi \frac{I}{2\pi r} \quad (r > a)$$

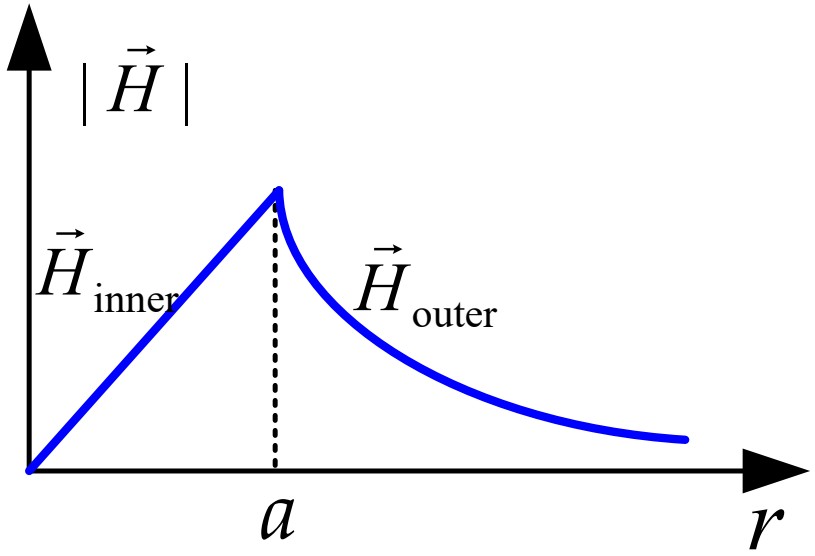
- Within the volume current: construct an auxiliary circle within the wire and perpendicular to the axis.




$$\oint_C \vec{H} \bullet d\vec{l} = \oint_C H_\varphi dl = H_\varphi \cdot 2\pi r = ?$$

$$I_{\text{ringed}} = J \cdot \pi r^2 = \frac{I}{\pi a^2} \cdot \pi r^2 = \frac{I r^2}{a^2}$$

$$\vec{H} = \frac{\vec{a}_\varphi I r}{2\pi a^2}$$




$$\vec{H}_{\text{inner}} = \frac{\vec{a}_{\varphi} I r}{2\pi a^2}$$

$$\vec{H}_{\text{outer}} = \vec{a}_{\varphi} \frac{I}{2\pi r}$$

# Example 1. Inner Self-Inductance

- Please determine the inner self-inductance at per unit length of a straight wire, of which the radius is *a*.
- From example 4 in § 5.1, we know in the wire…

$$\vec{H} = \frac{\vec{a}_\varphi Ir}{2\pi a^2}$$

- M-flux through the shadow area is

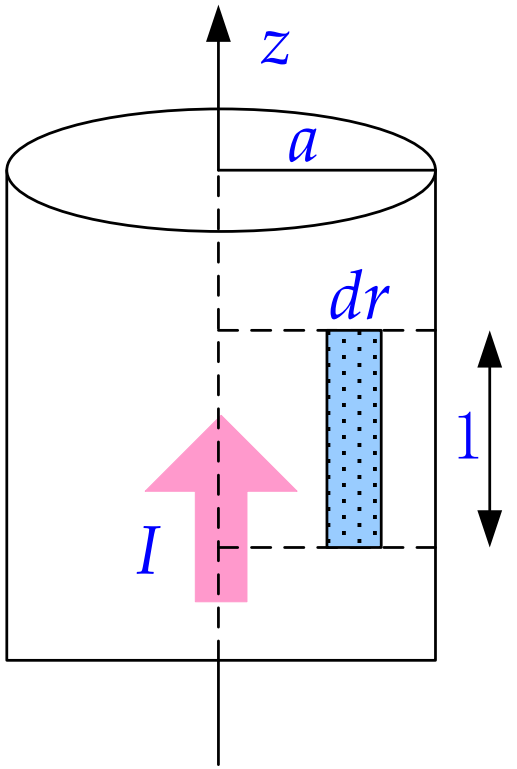
$$d\Phi = \vec{B} \bullet d\vec{S}_\varphi = \frac{\mu_0 Ir}{2\pi a^2} \cdot dr \cdot 1$$

- Current ringed by this portion of M-flux is
  - $I \times r^2/a^2$
- The corresponding M-flux-linkage is ?
- The overall M-flux-linkage is

$$\Psi = \int_0^a ?\, dr = ??$$

- The inner self-inductance is

$$L = \mu_0/8\pi$$

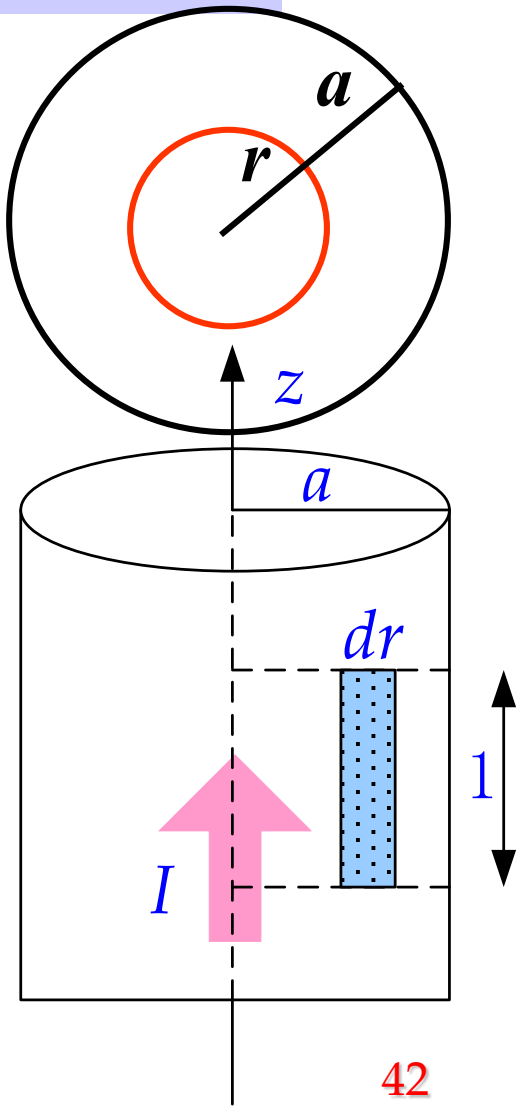


$$d\Psi = \frac{\pi r^2}{\pi a^2} \cdot d\Phi = \frac{\pi r^2}{\pi a^2} \cdot \left( \frac{\mu_0 \cdot I}{2\pi \cdot a^2} \right) \cdot r dr \quad 0 < r < a$$

$$\Psi = \int_0^a \frac{\mu_0 \cdot I}{2\pi \cdot a^4} \cdot r^3 dr = \frac{\mu_0 \cdot I}{8\pi}$$

$$L = \frac{\Psi}{I} = \frac{\mu_0}{8\pi}$$

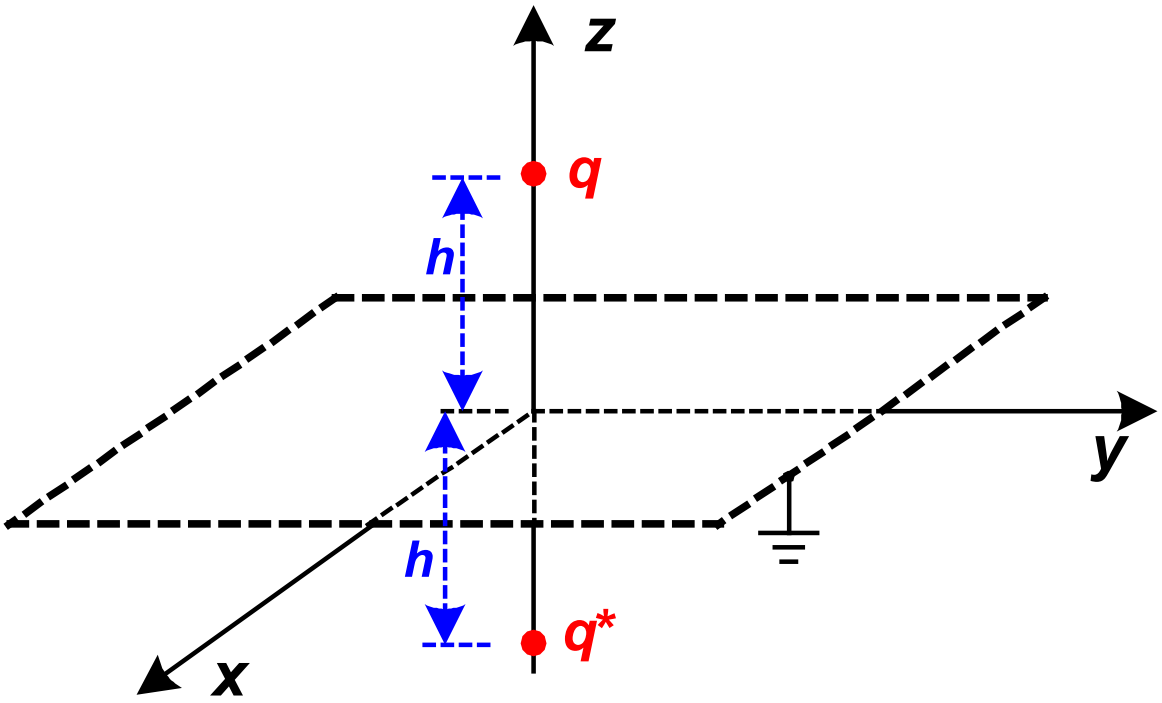
The inner self-inductance at per unit length of a straight wire.



# 第一类镜像法：平面镜像

(1)将导体移走,在“对称”点处放置一个“像”电荷$q^*$

(2)仍然要满足“导体板”存在时的条件,$q^*=?$




$$q^* = -q$$

# 小结：第二类“镜像”问题

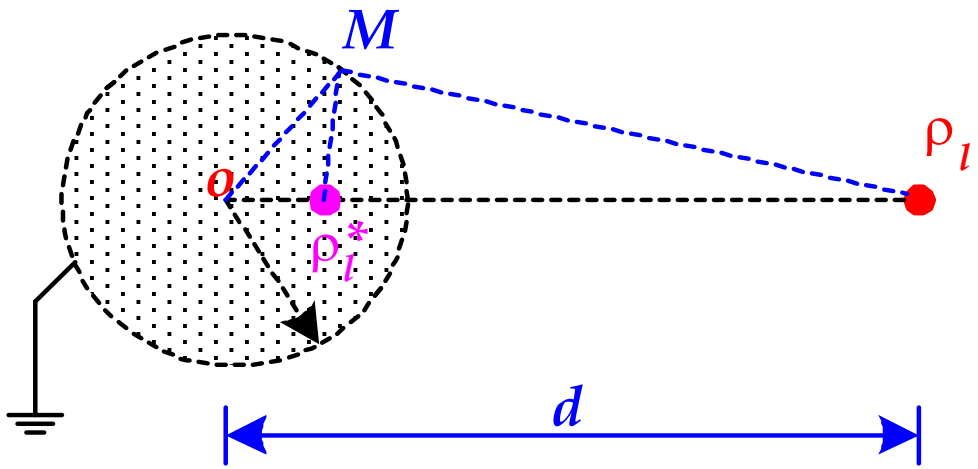
“线电荷”+“长导体柱”

所求场的区域中不能有镜像电荷！

镜像电荷-大小？-符号？-位置？

$$\rho_l^* = -\rho_l$$

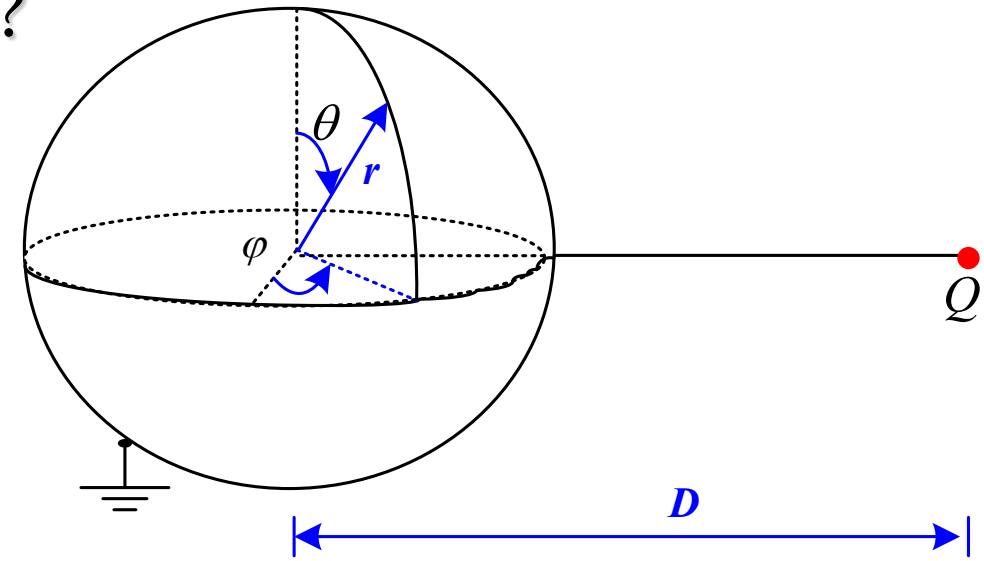
$$\left|OP^*\right| = \frac{a^2}{d}$$



# 第三类镜像法：球面镜像

例如：一个点电荷，位于接地导体球旁边，球半径为$a$

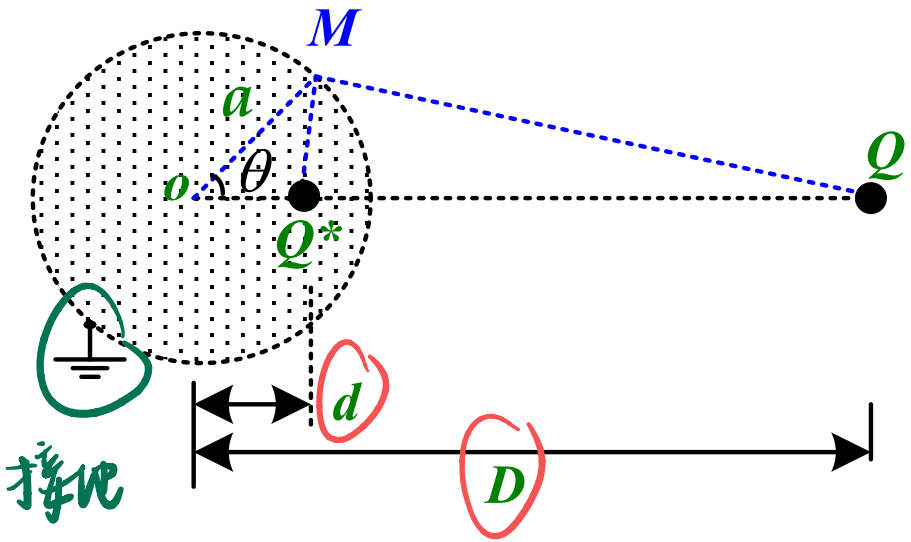
求：该电荷受到的静电力？



导体球上任一点$M$

$$\psi = \psi_q + \psi_{q^*}$$

$$= \frac{1}{4\pi \cdot \varepsilon_0} \cdot \left( \frac{Q}{r_{MQ}} + \frac{Q^*}{r_{MQ^*}} \right)$$




$$\because \psi = 0$$

$$\therefore \left\{ \begin{array}{l} \dfrac{Q}{r_{MQ}} + \dfrac{Q^*}{r_{MQ^*}} = 0 \\ r_{MQ}^{\ 2} = a^2 + D^2 - 2aD\cos\theta \\ r_{MQ^*}^{\ 2} = a^2 + d^2 - 2ad\cos\theta \end{array} \right.$$

$$\therefore \left\{ \begin{array}{l} Q^* = -\dfrac{a}{D} \cdot Q \\ |OQ^*| = d = \dfrac{a^2}{D} \end{array} \right.$$

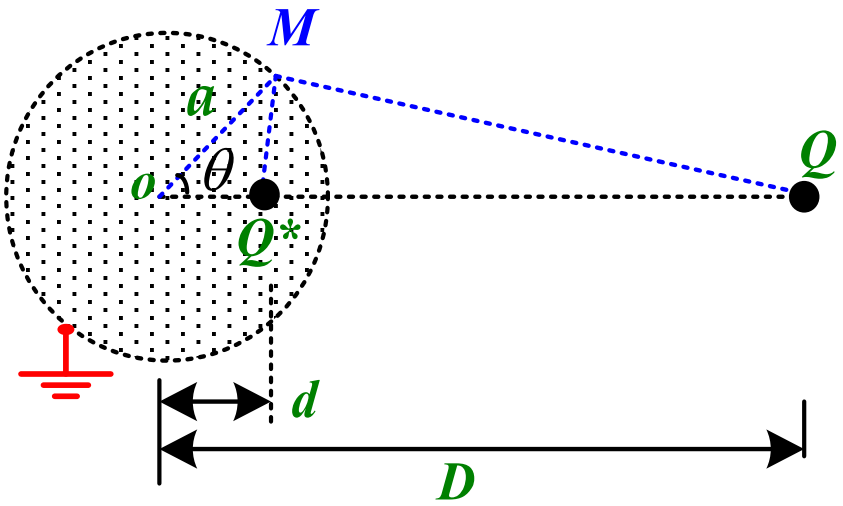
其他球面镜像问题：不接地导体球

掌握

O 导体球不带电

感应电荷可用镜像电荷来表征：

$$\begin{cases} Q^* = -Q \cdot a / D \\ d = a^2 / D \end{cases}$$




导体球总电量为0，因此在球心需要第2个镜像电荷 $-Q^*$ 来满足边界条件

O 导体球带电 $q_0$

感应电荷可用镜像电荷 $Q^*$ 来表征，为满足边界条件，在球心需要第2个镜像电荷 $Q''$：$Q^* + Q'' = q_0$